# QING LIU

Oort Fellow | Leiden Observatory, Leiden University

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## Research interests

Low surface brightness astronomy, with an emphasis on multi-wavelength characterization of Galactic cirrus and diffuse Galactic light; interstellar dust and ISM physics; intracluster light; astrostatistics and image analysis; astronomical instrumentation. Complementary interests include faint dwarf and ultra-diffuse galaxies, and galaxy assembly across environments.

## Employment

**Oort Fellow / Postdoctoral Researcher** 2024–present
Leiden Observatory, Leiden University, The Netherlands
Host: Prof. Henk Hoekstra

**Doctoral Researcher / Research Assistant** 2018–2024
Department of Astronomy & Astrophysics, Dunlap Institute for Astronomy & Astrophysics, University of Toronto, Canada

## Education

**Ph.D. in Astronomy & Astrophysics, University of Toronto** 2018–2024
Advisor: Prof. Roberto Abraham
Thesis: *Systematics Control in the Era of Ultra-Low Surface Brightness Imaging*.

**B.Sc. in Astronomy, University of Science and Technology of China** 2014–2018
Advisor: Prof. Xu Kong Co-advisor: Prof. Ralf Klessen (Heidelberg U.)
Thesis: *Parameter Estimation of Molecular Cloud–Star Cluster Evolutions on the BPT Diagram*.

## Survey & collaboration contributions

**Euclid — Local Universe Science Working Group (SWG-LU; WP-DET; WP-DIF). Milky Way & Resolved Stellar Populations Science Working Group (SWG-MWRSP).** Characterize Galactic cirrus in optical and near-infrared imaging; develop and validate foreground models for diffuse-light measurements; contribute to intracluster light and tidal stream studies.

**ARRAKIHS — MOC & DAT.** Contribute to Red Book sections on foreground systematics and target-selection strategies; simulate Galactic foregrounds for mock imaging datasets.

**Roman — SUPERBACK team.** Model the diffuse Galactic light component of the sky background for the Nancy Grace Roman Space Telescope.

**Dragonfly — Dragonfly UltraWide Survey (DFUWS).** Develop data-reduction and sky-background modeling methods that preserve diffuse emission, including wide-angle PSF characterization and optical cirrus separation.

## Technical skills

**Programming & research tools.** Python, R, C, IDL, SQL, and shell scripting; Git, Jupyter, and LaTeX.

**Statistical inference.** Bayesian parameter estimation; Markov chain Monte Carlo (MCMC; `emcee`); nested sampling (`dynesty`); gaussian process; covariance model and uncertainty estimation; machine learning.

**Imaging reduction & analysis.** Background and PSF modeling; source detection, aperture and isophotal photometry; astrometric and photometric calibration; dark and flat correction; imaging spectroscopic data reduction. Astropy, Photutils, SEP, astrometry.net, IRAF, SExtractor, SCAMP, and SWarp.

**Physical modeling.** Dust radiative transfer model; dust grain model (`dustem`); stellar-population synthesis (`fsps`); galaxy SED fitting; structural analysis on ISM; power spectrum analysis.

## Mentoring & teaching

### Student supervision and research mentoring

**Panagiota Ntova — Master thesis, Leiden University** 2025-2026
*Disentangling cirrus contamination with neural networks for Unbiased Characterization of Intra-cluster Light*.

**Liana Li — Ph.D. project (co-mentor), Universidade de São Paulo** 2025-2026
*Intracluster light in the S-PLUS and Dragonfly DFUWS survey*.

**Samir Singh — Undergraduate thesis, University of Toronto** 2022-2023
*Effects of Atmospheric Aerosol on Telescope Wide-angle Point Spread Function*.

**Rebecca Ceppas de Castro — Undergraduate thesis (co-mentor), University of Toronto** 2022-2023
*Searching for M33 satellite galaxies with deep Dragonfly imaging*.

### Teaching experience

**AST201: Stars and Galaxies — University of Toronto** Spring 2019–2022
Course enrollment: approximately 1,500 students.

**AST101: The Sun and Its Neighbours — University of Toronto** Autumn 2018
Course enrollment: approximately 1,500 students.

**AY14204: Galactic Astronomy — USTC** Spring 2018
Course enrollment: 40 students. Graduate level.

**02217001: Astronomical Labs — USTC** Autumn 2017
Course enrollment: 30 students. Software skills for data analysis.

## Observational & instrumental experience

**Dragonfly Telephoto Array — Operational observer** 2019–2024
Managed observing queues and project definitions; monitored robotic observations and troubleshot cameras, guiders, and focusers; coordinated with New Mexico Skies staff; maintained observing protocols and trained observers. Contributed to image diagnostics and pipeline development.

**Gemini North / GMOS — Principal investigator** 2022B
Follow-up imaging for the M33 dwarf satellite galaxy search.

**Hubble Space Telescope / ACS — Co-investigator** Cycle 31
*Two Extremely Faint Candidate Satellites of M33 Identified in Diffuse Light* (PI: E. Patel).

**Canada–France–Hawaii Telescope / SITELLE — Co-investigator** 2019B, 2020B, 2021A, 2022A
*Mapping H$\alpha$ Emission in Star-Forming Galaxies in Rich Galaxy Clusters* (PI: H. Yee).

**48-lens Dragonfly Telephoto Array.** Operation and maintenance, including image-quality control and guider, focuser, and photometric zero-point calibration.

**120-lens Dragonfly Spectral Line Mapper.** Construction contributions: camera and filter-tilter assembly, cabling, Arduino controller assembly, and focusing tests.

## Honors, awards & funding

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| Leids Kerkhoven-Bosscha Fonds (LKBF) grant | 2025 |
| Oort Postdoctoral Fellowship | 2024 |
| University of Toronto Thesis Research Award | 2023 |
| Ontario Graduate Scholarship for International Students | 2022–2023 |
| University of Toronto Program Level Fellowship | 2022 |
| Ontario Trillium Scholarship | 2018–2022 |
| Connaught Scholarship | 2018 |
| Best Undergraduate Thesis in Astronomy | 2018 |
| National Observatory of China Scholarship | 2017 |
| USTC Excellence Scholarship | 2015–2017 |

## Professional talks

**2026**

**Euclid Local Universe Meeting** Valladolid, Spain; November scheduled
*Invited talk — Galactic Cirrus in Euclid: Observations and Modelings*

**Physics seminar @ Kwansei Gakuin University** Hyogo, Japan; November scheduled
*Invited seminar — Unveiling the Faint Diffuse Universe using Telescopes from Ground and Space*

**TDLI Astronomy seminar** Shanghai, China; November scheduled
Galactic Cirrus in Euclid: Observations and Modelings

**The Olympian ISM Symposium** Paralia, Greece
Mapping the Diffuse ISM in the Milky Way with Deep Wide-Field Imaging

**Intra-Cluster Light @ Lorentz Center** Leiden, The Netherlands
*Invited review talk — The Diffuse Dust Foreground in Low Surface Brightness Imaging*

**Intra-Cluster Light @ Lorentz Center** Leiden, The Netherlands
Modeling and Disentangling Galactic Cirrus from Intracluster Light

**Shanghai Astronomical Observatory seminar** Shanghai, China
Low Surface Brightness Imaging: From Ground to Space

**USTC seminar** Hefei, China
Low Surface Brightness Imaging: From Ground to Space

**2025**

**Purple Mountain Observatory seminar** Nanjing, China
Revealing the Low Surface Brightness Universe from Ground to Space

**15th Cosmic Dust Meeting** Osaka, Japan
Diffuse Galactic Light Imaged by the Dragonfly Telephoto Array

**ISSI Euclid–ARRAKIHS Meeting** Bern, Switzerland
*Invited review talk — An Overview Talk on Galactic Cirrus*

**NOVA NW1 Meeting** Groningen, The Netherlands
Unveiling the Misty and Mysterious Low Surface Brightness Universe

**Department of Astronomy lunch talk, Tsinghua University** Beijing, China
The Misty and Mysterious Low Surface Brightness Universe Imaged by Dragonfly

**National Astronomical Observatories seminar** Beijing, China
The Misty and Mysterious Low Surface Brightness Universe Imaged by Dragonfly

**2024–2018**

**244th AAS Meeting** Madison, WI, USA; 2024
Diffuse Optical Galactic Cirrus in Dragonfly's Deep Imaging

**Diffuse Cosmic Background and Low Surface Brightness Universe** Aspen, CO, USA; 2024
Optical Galactic Cirrus in Deep Wide-Field Imaging from Dragonfly

**CASCA Annual General Meeting** Penticton, BC, Canada; 2023
Controlling Background Systematics in Low Surface Brightness Imaging Using Dragonfly

**CITA Astro-coffee** Toronto, ON, Canada; 2023
A Recipe for Unbiased Background Modeling in Deep Wide-Field Astronomical Images

**Rubin–DECaLS–HSC Joint Group Seminar** Princeton University, NJ, USA; 2022
*Invited seminar — Controlling Background Systematics in Low Surface Brightness Imaging*

**Galaxy Journal Club** University of Toronto, Canada; 2021
A Method to Characterize the Wide-angle Point-Spread Function of Astronomical Images

**Astro-ph talk** University of Toronto, Canada; 2021
Wide-angle PSF Modeling in Deep Wide-Field Images

**Galaxy Day** University of Toronto, Canada; 2019
SITELLE H$\alpha$ Imaging Spectroscopy of Emission-Line Galaxies in Clusters at $z \sim 0.25$

**Astro-lunch** Shanghai Normal University, China; 2018
The Effect of Star-Formation Histories on Stellar-Population Properties in SED Fitting

**CANDELS Collaboration Meeting** University of California, Santa Cruz, USA; 2018
Properties of Stellar Populations in Star-Forming Galaxies Based on SED Fitting

## Publication list (selected)

See my ADS Library for the full publication list.

### Manuscripts in preparation

**Liu, Q.**, et al. Galactic cirrus in the Perseus Cluster field: A panchromatic view with Euclid, CFHT, GALEX, IRIS, and Planck.. *In preparation.*

### First- and second-author refereed papers

1. **Liu, Q.**, Hoekstra, H., Ellien, A. (2026).
Statistical modeling of foreground Galactic cirrus for diffuse halo light measurement. submitted to *A&A*
2. **Liu, Q.**, Martin, P. G., Abraham, R. G., et al. (2025).
Quantitative morphology of galactic cirrus in deep optical imaging: Statistical structural analysis in a multiwavelength perspective. *A&A*, **704**, A269.
3. Li, D., **Liu, Q.**, Eadie, G. M., et al. (2025).
Candidate Dark Galaxy-2: Validation and Analysis of an Almost Dark Galaxy in the Perseus Cluster. *ApJL*, **986**, L18.
4. **Liu, Q.**, Abraham, R. G., Martin, P. G., et al. (2025).
Fuzzy Galaxies or Cirrus? Decomposition of Galactic Cirrus in Deep Wide-field Images. *ApJ*, **979**, 175.
5. **Liu, Q.**, Abraham, R. G., Martin, P. G., et al. (2023).
A Recipe for Unbiased Background Modeling in Deep Wide-field Astronomical Images. *ApJ*, **953**, 7.
6. **Liu, Q.**, Abraham, R. G., Gilhuly, C., et al. (2022).
A Method to Characterize the Wide-angle Point-Spread Function of Astronomical Images. *ApJ*, **925**, 219.
7. **Liu, Q.**, Yee, H. K. C., Drissen, L., et al. (2021).
SITELLE H$\alpha$ Imaging Spectroscopy of $z \sim 0.25$ Clusters: Emission-line Galaxy Detection and Ionized Gas Offset in Abell 2390 and Abell 2465. *ApJ*, **908**, 228.
8. **Liu, Q.**, Wang, E., et al. (2018).
Elevation or Suppression? The Resolved Star Formation Main Sequence of Galaxies with Two Different Assembly Modes. *ApJ*, **857**, 17.

### Contributed refereed papers (selected)

9. Euclid Collaboration: Fu, Y., Bouwens, R., Caputi, K. I., et al. (including **Liu, Q.**) (2026).
Euclid Quick Data Release (Q1). XXXVIII. Euclid spectroscopy of quasars. 1. Identification and redshift determination of 3500 bright quasars. *A&A*, **711**, A285.
10. Euclid Collaboration: Urbano, M., Duc, P.-A., Saifollahi, T., et al. (including **Liu, Q.**) (2025).
Euclid: Early Release Observations of diffuse stellar structures and globular clusters as probes of the mass assembly of galaxies in the Dorado group. *A&A*, **700**, A104.
11. Pasha, I., van Dokkum, P. G., **Liu, Q.**, et al. (2025).
The Bullseye: HST, Keck/KCWI, and Dragonfly Characterization of a Giant Nine-ringed Galaxy. *ApJL*, **980**, L3.
12. Shen, Z., Bowman, W. P., van Dokkum, P., et al. (including **Liu, Q.**) (2024).
First Results from the Dragonfly Ultrawide Survey: The Largest Eleven Quenched Diffuse Dwarf Galaxies in 3100 deg$^2$ with Spectroscopic Confirmation. *ApJ*, **976**, 75.
13. Gilhuly, C., et al. (including **Liu, Q.**) (2020).
The Dragonfly Edge-on Galaxies Survey: Shaping the Outer Disk of NGC 4565 via Accretion. *ApJ*, **897**, 108.
14. Danieli, S., et al. (including **Liu, Q.**) (2020).
The Dragonfly Wide Field Survey. I. Telescope, Survey Design, and Data Characterization. *ApJ*, **894**, 119.

## Professional service & outreach

**Peer review.** Referee for *The Astrophysical Journal* and *Astronomy & Astrophysics* (2021–present); Euclid internal publication referee for Local Universe (SWG-LU) and Galaxy Evolution (SWG-GAEV).

**Public engagement.** University of Toronto Amateur Astronomers Association (2018–2020): co-organized meteor-shower trips and supported stargazing and astrophotography. AstroTour volunteer (2019–2020).

**Campus observing.** Assisted student observing projects at the U of T campus observatory (2018).